SPY HUNTER

G-6155 · INTERCEPTOR ARCADE TRIBUTE

RUST + BEVY 0.20 · ORIGINAL PIXEL ART



MISSION DEBRIEF · NOVEMBER 2026

orbitmechanic.github.io/spy-hunter-web

A playable Rust + Bevy tribute, built and shipped in one session.

From a 1983 arcade memory to a shipped browser game

The assignment was deliberately broad: research the original, build an original game in the same spirit, generate the art and audio, test it, publish it, and dock it on the station.

RESEARCH

Deconstruct Spy Hunter's top-down chase, weapons van, civilian rule, car-to-boat switch, and Peter Gunn-era sound.

FOUNDATION

New project at games/bevy/spy-hunter, Cargo + Bevy 0.20.1, public GitHub repo from the first commit.

ASSETS

Original Krita pixel art, a 5×7 pixel TTF, a 23.7 s chiptune loop, and ten SFX — no ripped assets.

SHIP

Native + wasm builds, screenshot harness, GitHub Pages deployment, landing-page dock, and this deck.

4
TERRAINS

23.7 s
ORIGINAL LOOP

6 / 6
SIM TESTS

0
RIPPED ASSETS

The target: Spy Hunter, Bally Midway, 1983

A vertical-scrolling vehicular-combat game. The player drives the G-6155 Interceptor, destroys enemy traffic, protects civilians, and collects special weapons from a van. The road forks into new weather and terrain; at water, the car becomes a speedboat.

GAMEPLAY

Top-down, 4-lane road, machine gun, oil / smoke / missiles, helicopter bombs, off-road and civilian penalties.

LOOK

Chunky sprite cars, hard lane lines, bold roadside colour blocks, a compact dashboard HUD.

SOUND

An electronic arrangement of Henry Mancini's Peter Gunn theme; engine and weapon cues under it.

RULES

Score accrues while moving on-road. Civilian hits freeze scoring. Extra cars at score thresholds. No ending.



The Peter Gunn theme is copyrighted. This tribute writes an original spy-surf cue instead of reproducing it.

One Interceptor, four terrains, endless pursuit

01
DRIVE & SURVIVE

Curving four-lane highway, coast / snow / water / night cycle, off-road timer, craters, helicopter bombs, extra lives.

02
FIGHT

Hold-fire machine gun, oil slick, smoke screen, homing missile, weapons van, switchblade cars, pursuing boats.

03
ARCADE SHELL

Title, HUD, pause veil, game-over stats, hi-score, touch controls, and a browser-first canvas.





Delta Waterway: car becomes boat, oil becomes flamethrower.

A small ECS with a very clear loop

Bevy 0.20.1 · edition 2024 · fixed 960 × 720 logical canvas · one 2D camera

TITLE → **IN GAME** → **GAME OVER**

Enter starts a run; death with zero cars ends it; Enter redeploys.

INPUT | ROAD | SPAWN | MOVE | FIRE | COLLIDE | HUD

Update systems run in a chain; gameplay pauses with one resource flag.

Game
score · lives · terrain · timers

Weapons
oil · smoke · missiles · van

Road
scroll distance

Rng
deterministic xorshift

Fx
shake + particles

CONTENT PIPELINE

Krita 6
PNG sprites
+ terrain

→

make_font.py
5×7 pixel
TTF

→

make_audio.py
WAV music
+ 10 SFX

→

wasm-bindgen
WebGL 2
bundle

Original pixel art, drawn through Krita's Python API

PALETTE

Twelve fixed colours: ink, asphalt, lane cream, grass, snow, water, body, enemy red, blue, civilian teal, gold, ice.

SPRITES

Cars, boats, helicopter, van, pickups, explosions, trees, rocks, buoys, buildings, icons, and a 21-sprite editable board.

TERRAIN

Four 960×720 fields: grass, snow, water, night city. The four-lane road itself is built from recycled Bevy sprites.

FIX

Krita creates an opaque white Background layer. The export script hides it before writing pixels so alpha survives.







An original spy-surf cue, synthesized from scratch



FORM
8-bar loop at 240 BPM in E minor. Walking square bass, tremolo square lead, triangle pad, noise drums.

SFX
Ten cues: machine gun, enemy shot, explosion, missile, pickup, civilian penalty, death, extra life, terrain, UI.

FORMAT
22.05 kHz mono 16-bit WAV; one 1.0 MB music loop, 1.2 MB total audio. No samples from the arcade machine.

WHY ORIGINAL
Peter Gunn is owned by its composers/publishers. A bespoke cue keeps the tribute legally clean and gives the game its own identity.

Compiling is not proof. Every claim has a check.



TITLE



HIGHWAY



ALPINE PASS



GAME OVER

cargo test: 6/6 pass · cargo check: clean · native capture at frames 120–520 · web smoke test in headless Chromium

Four bugs worth remembering

SCORE STUCK AT 4

Score was added as (speed × dt × 0.11) as u64. At 60 fps the fraction truncated to zero every frame. Fix: keep a fractional accumulator and add whole points only.

BEVY B0001

Two queries touched Transform — one mutably — without disjoint filters, so the app panicked on the first frame. Fix: Without<Player/Enemy/Missile> filters.

WHITE SPRITE CARDS

Every Krita export had an opaque white background because createDocument adds a Background layer. Fix: hide that layer before merge-exporting.

WEB AUDIO + SIZE

Browsers suspend AudioContext until a trusted gesture, and the raw wasm was over 100 MB. Fix: capture/resume contexts in the shell, wasm-bindgen + wasm-opt -Oz.

Source, static build, Pages, and a new docking bay



The game repo carries source, art boards, tests, and the build/deploy scripts.

The web repo carries only the staged runtime: index.html, CSS, JS glue, wasm, assets, .nojekyll.

LIVE URLS

```
PLAY     https://orbitmechanic.github.io/spy-hunter-web/
SOURCE   https://github.com/orbitmechanic/spy-hunter
BUILD    https://github.com/orbitmechanic/spy-hunter-web
STATION  https://orbitmechanic.github.io/ (Bay 08)
```

What I would build next, in order

01 GAMEPAD + MOBILE POLISH

Wire gamepad axes and tune the touch hit targets. Add haptics and a rotate-to-landscape hint.

02 MORE ROAD STORIES

Branching forks, a boathouse choice, motorcycles, limousines, and a second helicopter pattern.

03 RENDER BUDGET

Bake the sprite set into an atlas and batch the 100+ road parts. Target a stable 60 fps on integrated GPUs.

04 PERSISTENCE

Local high-score table and per-terrain bests; optional serverless leaderboard.

05 AUDIO MIX

Duck the music under explosions, add engine pitch tied to speed, and a mute toggle on the title screen.

06 ACCESSIBILITY

Remap keys, reduce screen shake, high-contrast HUD, and a slower starting difficulty.

A new arcade cabinet is docked at the station.

Play it: orbitmechanic.github.io/spy-hunter-web
Source: github.com/orbitmechanic/spy-hunter

Original pixel art · original chiptune · Rust + Bevy 0.20.1 · GitHub Pages





Original work. No ROM, cabinet, or arcade asset is included.